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THERMAL MODELING OF A GEOTHERMAL WELL FIELD FOR GROUND SOURCE WATER HEAT PUMP SYSTEMS

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ABSTRACT

Heat pumps are mechanical systems that provide heating to a space in the winter, and cooling in the summer. They are increasingly popular because the same system provides both cooling modes, depending on the direction of the cycle upon which they operate. For proper operation, the heat pump must be connected to a constant temperature thermal reservoir which in traditional systems is the ambient air. In ground source heat pumps however, subterranean ground water is used as the thermal reservoir. To access the subterranean groundwater, "geothermal" wells are drilled into the formation. Water from the building heating or cooling system is circulated through the wells thereby promoting heat exchange between the coolant water and the subterranean formation. The potential for higher efficiency heating and cooling has increased the utilization of ground source HVAC systems. In addition, their compatibility with a naturally occurring and very stable thermal reservoir has increased their use in the design of sustainable or green buildings and man-made environments. This paper is concerned with the development of a mathematical model describing the thermal processes that occur in a circulating geothermal well as the water thermally interacts with the surrounding formation. First principles are utilized to develop a general single well model that is able to predict the heating or cooling of a well with time. The single well model is then utilized to build a simulation for an infinite line of adjacent wells that thermally

interact with each other over time. The behavior of a single isolated well and a line of evenly spaced wells is discussed and compared with recent experimental data for circulation in an isolated well.

INTRODUCTION

Geothermal wells are of great interest for the heating and cooling industry for potential increases in efficiency associated with using the ground as a thermal reservoir instead of the atmosphere. A typical geothermal well is between 400-1500 feet deep and six inches in diameter. There are many types of geothermal wells used in ground source heat pumps. The geothermal wells studied in this paper are referred to as standing column wells (SCW).

Standing column wells are open loop and physically interact with the ground and groundwater. Water is extracted from below the water table into a building, run through a heat exchanger, and returned to the ground. The ground around the well is specific to the area and ranges from porous to solid rock. Porous rock allows for advection through the ground formation while solid rock does not allow for advection. However, solid rock has random cracks which allow for water flow away from the well. Hence, the ground formation properties are anisotropic. Since the ground formation is a semi-infinite solid, the temperature profile is transient in both space and time.

Heat pumps are mechanical systems that run a thermodynamic cycle to pump heat from low temperature to high temperature. The advantage of using ground source heat pumps is associated with the stability of ground water temperatures throughout the year compared with surface air temperatures used in air source heat pumps. The physical limitations of the wells are determined by the water properties. During the heating season, the groundwater is cooled by the heat pump and approaches the freezing temperature of water which affects the submersible pumps' operation. This problem arises when a heat pump system capacity exceeds the physical limits of the geothermal well, and is a primary driver for accurate computational models.

Geothermal wells have been researched extensively for surface water and ground-coupled (closed loop) geothermal systems, and design manuals such as the ASHRAE Commercial/Industrial Ground-source Heat Pump Engineering Manual [1], and Ground-Source Heat Pumps: Design of Geothermal Systems for Commercial and Institutional Buildings [2] have been produced. SCWs have attracted more attention in previous years; Rees et al [3], Deng et al [4] and Spitler et al [5] focused on single SCW well modeling using both analytical and numerical techniques. The defining properties that limit operation as noted in these investigations are the ground thermal conductivity, bleed rate, and depth. However, these characteristics affect the near well fluid mechanics and heat transfer, specifically advection through the ground formation due to pressure gradients and porous media flow through the formation. The current study probes the farfield interactions between wells for the design of well fields for large installations.

Since geothermal wells have limited depths, the thermal capacity of a single well is finite and depends on the properties of the rock and the aquifer surrounding the well. Large building loads typically exceed single well limits and require multiple wells. This paper presents the development of a numerical model for determining multiple well thermal interactions using "effective" properties of the ground derived by experimentation. First, an axisymmetric numerical model is developed and compared with empirical data to assess the transient behavior of a single well. Second, a multiple well model is developed to assess the effect of operating multiple wells concurrently at varying inter-well distances. Finally, the multiple well model is used to verify the common practice separation distance of approximately ten meters for geothermal well fields.

MODEL DEVELOPMENT

The SCW temperature profile is influenced by a thermal load entering at the top of the well. The thermal load is transferred to the surrounding formation through conduction and advection, and by conduction to return water pumped up the middle of the geothermal well in two high density polyethylene (HDPE) pipes.

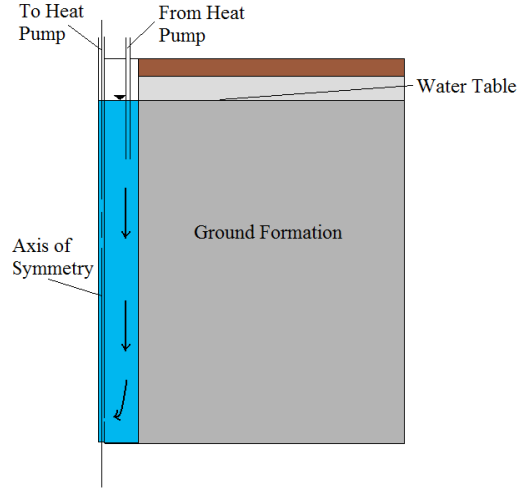


Fig. 1: Diagram of standing column well operation

The ground formation spatial domain is a semi-infinite solid with properties that are anisotropic due to porosity, and random cracks. Figure 1 depicts the problem domain.

For analysis, the porosity, and random non-homogeneities are assumed to be equal in magnitude and direction, and are lumped together with advection producing effective properties for the ground formation such as an effective conductivity, k_{eff} , and effective diffusivity, α_{eff} . For this study, effective properties are determined through empirical data supplied by ARB Geowell, the operator.

The energy balance for the water flowing down the well is derived by integration in the radial direction from the outer wall of the polypipe (R_{po}) tube to the wall of the well (R_o) and is given by

$$\frac{\partial T_w(z,t)}{\partial t} = -\frac{2\pi h_o R_o}{\rho_w C_{pw} A_{cd}} (T_w(z,t) - T_g(0,z,t)) - 2 \frac{(T_w(z,t) - T_r(z,t))}{R_{pipe}} - \frac{\dot{m}}{\rho_w A_{cd}} \frac{\partial T_w(z,t)}{\partial z} \quad (1)$$

The flow is one-dimensional axial transient flow with two Robin boundary conditions located in Eq. (1) as the first two terms on the right hand side of the equation. The flow is assumed to be fully developed turbulent flow. The boundary conditions in the axial direction are given by

$$T_w(0,t) = T_r(0,t) + \frac{\dot{Q}(t)}{m C_{pw}} \quad (2)$$

at the top of the well where $\dot{Q}(t)$ is the building heat load with time, and

$$T_w(H,t) = T_r(H,t) \quad (3)$$

at the bottom of the well. The value of R_{pipe} is given by

$$R_{pipe} = \frac{1}{\pi h_{ip} D_{ip}} + \frac{\ln\left(\frac{D_{op}}{D_{ip}}\right)}{2\pi k_p} + \frac{1}{\pi h_{op} D_{op}} \quad (4)$$

The energy balance for the upcoming flow is derived in the same fashion as Eq. (1) in terms of the mixed mean fluid temperature within the polypipe inner radius (R_{pi}) leading to a one-dimensional axial transient flow governing equation given by

$$\frac{\partial T_r(z, t)}{\partial t} = \frac{2(T_w(z, t) - T_r(z, t))}{R_{pipe}} - \frac{m}{\rho_w A_{cu}} \frac{\partial T_r(z, t)}{\partial z} \quad (5)$$

Again, the flow is one-dimensional axial transient flow with a Robin boundary condition at the wall, given in Eq. (5) by the first term on the right hand side, and a symmetry boundary condition at the axis of symmetry. The axial boundary conditions are the same as the annular flow.

The energy balance derived for the ground is a one-dimensional transient model given by

$$\frac{\partial T_g(r, z, t)}{\partial t} = \alpha_g \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T_g(r, z, t)}{\partial r} \right) \quad (6)$$

The conduction in the axial direction is neglected due to the aspect ratio (R_{therm}/H) of the problem. The thermal extent in the radial direction, R_{therm} , is of the order 10 feet, and the depth of the well, H , is of order 100 feet. Therefore, by normalizing the energy equation conduction in the radial direction is dominant and Eq. (6) is produced.

Since the flow is turbulent in the pipe, the convective heat transfer coefficients were bounded using the Gnielinski equation, the Karman-Nikuradse equation, and Table 14-3 from Kays and Crawford [6] for fully developed turbulent flow between concentric circular-tube annuli. The Gnielinski equation correlates over Prandtl numbers of 0.5 to 2000, and Reynolds numbers of 2300 to 5×10^6 and is given by

$$Nu_D = \frac{(Re_D - 1000) Pr^{c_f/2}}{1.0 + 12.7 \sqrt{c_f/2} \left(Pr^{2/3} - 1.0 \right)} \quad (7)$$

The Karman-Nikuradse equation correlates well with empirical data over Prandtl numbers of 0.6 to 6, and is given by

$$Nu_D = \frac{Re_D Pr^{c_f/2}}{0.88 + 13.39 \left(Pr^{2/3} - 0.78 \right) \sqrt{c_f/2}} \quad (8)$$

The Prandtl number range for this problem is 4 to 10. Therefore, the Karman-Nikuradse equation is not very precise but offers an estimate of the heat transfer coefficient for comparison.

The heat transfer coefficient range for the wall of the well (h_o), the outer wall of the polypipe (h_{op}), and the inner wall of the polypipe (h_{ip}) were 540-1500 W/m²K, 4000-5000 W/m²K, and 7800-10000 W/m²K, respectively, for the flow parameters used in the empirical data collected. Although the ranges were large for these values, the dominant resistances to heat transfer were the conduction through the ground formation, and the pipe conduction through the HDPE pipe wall. Hence, the convective resistances have a small influence on the thermal dynamics of the problem and any convective coefficient in these ranges produces similar results.

Axisymmetric Numerical Model

An axisymmetric numerical solution was performed on the wellbore to allow for long time calculations. This model is derived by discretizing Eq. (1), (5), and (6) generating, respectively, equations (9), (10), and (11)

$$T_w(z, t + \Delta t) = (1 - Co_d - NTU_{1d} - NTU_{2d}) T_w(z, t) + NTU_{1d} T_g(z, t) + NTU_{2d} T_r(z, t) + Co_d T_w(z - \Delta z, t) \quad (9)$$

$$T_r(z, t + \Delta t) = (1 - Co_u - NTU_{1u}) T_r(z, t) + NTU_{1u} T_w(z, t) + Co_u T_r(z + \Delta z, t) \quad (10)$$

$$T_g(r, z, t + \Delta t) = (1 - 2Fo) T_g(r, z, t) + (Fo - \frac{\alpha_g \Delta t}{2r \Delta r}) T_g(r - \Delta r, z, t) + (Fo + \frac{\alpha_g \Delta t}{2r \Delta r}) T_g(r + \Delta r, z, t) \quad (11)$$

where $z = 0$ is the top of the well, and Co_d , NTU_{1d} , NTU_{2d} , Co_u , NTU_{1u} , and Fo are given by

$$Co_d = \frac{m \Delta t}{\rho_w A_{cd} \Delta z} \quad (12)$$

$$NTU_{1d} = \frac{2\pi R_o h_o \Delta t}{\rho_w C_{pw} A_{cd}} \quad (13)$$

$$NTU_{2d} = \frac{2\Delta t}{\rho_w A_{cd} C_{pw} R_{pipe}} \quad (14)$$

$$Co_u = \frac{m \Delta t}{\rho_w A_{cu} \Delta z} \quad (15)$$

$$NTU_{lu} = \frac{2\Delta t}{\rho_w A_{cu} C_{pw} R_{pipe}} \quad (16)$$

$$Fo = \frac{\alpha_g \Delta t}{\Delta r^2} \quad (17)$$

This model allows for solution of the temperature variation in the ground formation, and in the well for long time periods of order days, weeks or months. Care was taken in the discretization of the ground formation due to large temperature gradients in the ground formation at $r = 0$, and in the well annulus at $z = 0$. Initial discretization units were 1 meter increments in the z -direction, and 0.0013 meter increments in the radial direction. As the radius and depth of well become larger, fine grids are unnecessary and increase computation time with little to no benefit. Hence, an adaptive grid was utilized in both the axial and radial direction. The time step was calculated using the first components on the right hand side of Eqs. (9), (10), and (11). Each of these components must remain above zero for the solution to converge. The time step used in the current analysis with the experimental properties was 0.44s. One year of operation was modeled in approximately fourteen days.

Multiple Well Cartesian Numerical Model

Since a single well's thermal capacity is limited and is not sufficient for large building loads, a field of wells is utilized to increase capacity. The influence of these wells on each other is important because thermal encroachment on a well by another lowers the efficiency of the entire field due to thermal saturation, thereby increasing the number of wells needed to supply a sufficient load for the building.

To develop the model, the same numerical solution is used for the wellbore as for the axisymmetric model but the ground formation energy equation and the discretized ground formation energy equation, respectively, are given by

$$\frac{\partial T_g(x, y, z, t)}{\partial t} = \alpha_g \left(\frac{\partial^2 T_g(x, y, z, t)}{\partial x^2} + \frac{\partial^2 T_g(x, y, z, t)}{\partial y^2} \right) \quad (18)$$

$$T_g(x, y, z, t + \Delta t) = (1 - 4Fo)T_g(x, y, z, t) + Fo(T_g(x + \Delta x, y, z, t) + T_g(x - \Delta x, y, z, t) + T_g(x, y + \Delta y, z, t) + T_g(x, y - \Delta y, z, t)) \quad (19)$$

Using Cartesian coordinates instead of cylindrical coordinates causes some numerical problems in the analysis of the ground formation, specifically in the interaction of the ground formation with the well. To link the well water temperatures to the ground, the circular well was assumed to be square in the computational domain of the ground as shown in Fig. 3. Figure 3 also shows the grid for the computational domain. The grid is adaptive and changes with distance from the well. A similar

analysis was used as in the axisymmetric model whereby the first components of the right hand side of Eqs. (9), (10), and (19) must be above zero for convergence. For the well, the cross-sectional area was maintained to provide the same Courant numbers for the Cartesian model as in the axisymmetric model. The defining thermal resistance in the model is the ground formation, therefore changing the convection resistance from the water in the annulus to the ground formation has little effect on the problem and produces the same results as in the axisymmetric problem. This assumption was verified in the Results section.

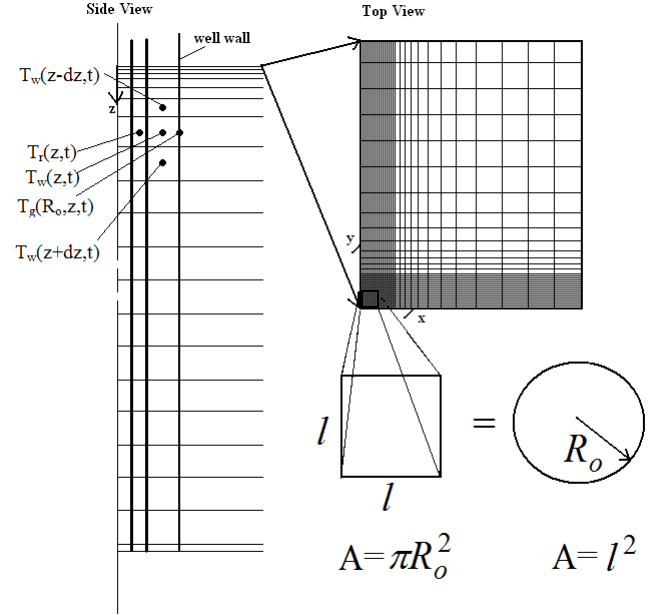


Fig. 3: The integration of a square well to simulate the circular well bore with a general grid for the computational domain. Each z -plane is discretized in the x and y directions. The nodes inside the square well are equal to the water temperature of the well at each z .

The NTU_{ld} , and the Fourier number for the ground formation are given, respectively, by

$$NTU_{ld} = \frac{4lh_o\Delta t}{\rho_w C_{pw} A_{cd}} \quad (20)$$

$$Fo = \frac{\alpha_g \Delta t}{\Delta L^2}, \Delta L = \Delta x = \Delta y \quad (21)$$

The multiple well Cartesian numerical model allows for thermal interaction between wells spaced a distance apart, as well as the ability to assess the effects of a line of wells running concurrently. The hypothesis is that a line of wells spaced an infinite distance apart will perform similarly to the single well model, shown in the Results section, and will deviate from that performance as the distance between the wells approaches $2R_o$. At a distance, $2R_o$, the problem becomes a two dimensional

problem in y and z with a wall source at $t = 0$. Figure 4 shows a diagram of an infinite row of wells operating concurrently. The domain space being analyzed is given by the gray area. The method of images is used to decrease the computational domain size.

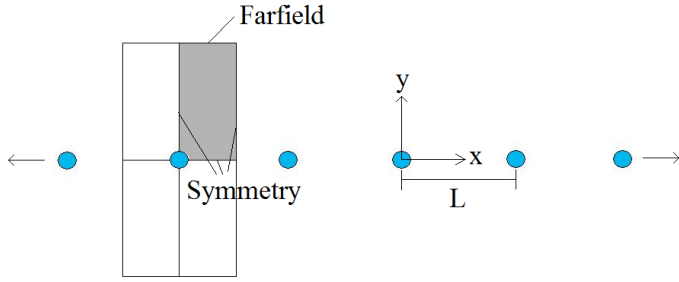


Fig. 4: A row of wells running co-currently.

The gray area is the computational domain defined by three symmetry boundary conditions and one farfield boundary condition.

Non-uniform grid spacing, with fine spacing near the well, and coarse spacing far from the well, allowed for lower computational time. The non-uniform grid used a modified form of Eq. (21) given by

$$Fo = \frac{\alpha_g \Delta t}{\Delta L_1 \left(\frac{\Delta L_1 + \Delta L_2}{2} \right)} \quad (22)$$

where ΔL_1 is the length between the base node and an adjacent node, and $(\Delta L_1 + \Delta L_2)/2$ is a length defining the total length between boundaries of nodes.

EMPIRICAL DATA COLLECTION

An experiment was performed by ARB Geowell on a geothermal well on Villanova University's campus used to provide heating and cooling to a residence hall. The well is 800 ft (243 m) deep and 6 in. (0.152 m) in diameter. It was installed on January 5, 2009 and data was taken on January 8th–11th, 2009. The flow rate was 32 GPM (2.0186 kg/s), heating load was 213,000 BTUH (63 kW), and pump power input was 1.5 hp (1.12 kW). The water table is 70 feet down hole and the ground composition is predominantly Gneiss. The experiment showed the ground to have an effective conductivity of 1.856 BTUH/ft°F (3.2 W/mK) and a thermal diffusivity of 1.06 ft²/day (1.14·10⁻⁶ m²/s). The effective conductivity is calculated using a linear regression estimation method from Witte et al [7]. Density and specific heat values for rock are from Incropera and Dewitt [8].

RESULTS

Model Validation

The models were validated by using the operating conditions of the experiment on Villanova's campus and the thermal properties derived from the experiment. In these experiments and simulations, the water-filled well is initially at the ground temperature. At $t = 0$, hot water is circulated into the well while the inlet and return water temperatures are monitored. The results for both models are shown in Fig. 5. The convection heat transfer coefficients, h_o , h_{op} , and h_{ip} allow for a $\pm 1^\circ\text{C}$ variation in the results. The axisymmetric data was fit to the empirical data by varying these parameters because of short computational times. The multiple well model was not fit to the data using this method due to the constraints on computational time.

In Fig. 5, the model has a transition during the first hour. The model is dominated by well heating during the first half hour. This region has a slope of zero when a polypipe material having an infinite resistance is used. The extent of this region is determined by the time it takes for the volume of water in the well to run a full circuit through the heat pump system. In non-dimensional terms, this requires

$$Co_u + Co_d = 1 \quad (23)$$

Or alternatively, the total volume of the well minus the volume of the polypipe divided by the volumetric flow rate defines a characteristic heatup time given by:

$$t_{\text{heatup}} = \left(\frac{\rho V_{\text{pipe}} + \rho V_{\text{annulus}}}{\dot{m}} \right) = 0.56 \text{ hr} \quad (24)$$

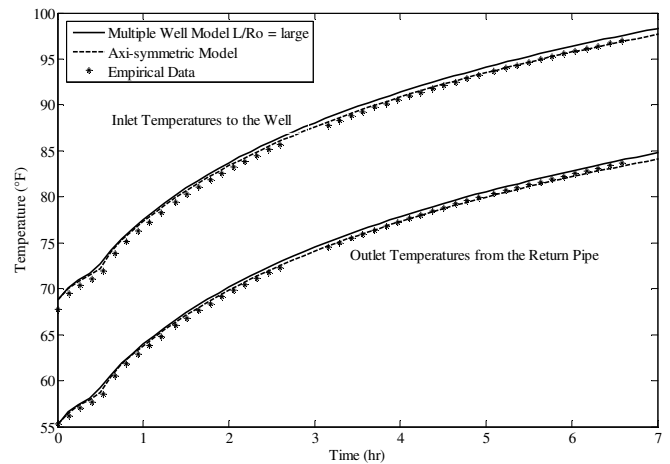


Fig. 5: Validation data for the multiple well model and the axisymmetric model using empirical data

The second region of the model is dominated by radial conduction where the temperature profile is given by

$$T_g(r, t) = f(e^{-t}) \quad (25)$$

Therefore, plotting in semi-logarithmic form provides a log-linear profile as time becomes large as shown in Fig. 6.

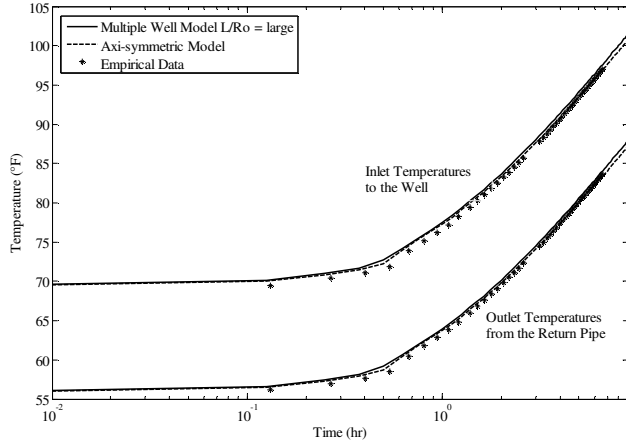


Fig. 6: Validation data in semi-logarithmic form depicting linear profile due to radial conduction.

Multiple Well Model Results

The multiple well model was run six times using values of $L/Ro = 2.2, 4, 8, 16, 32,$ and 64 . The radius of the well was used to non-dimensionalize the inter-well distance. This corresponds to a maximum distance between wells of 4.7m (15.5ft). Figure 7 shows the inlet temperatures produced by the simulations.

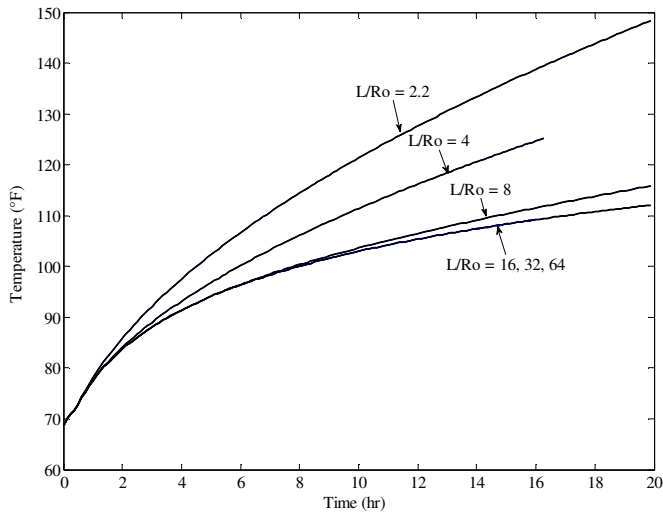


Fig. 7: Multiple well model results showing the deviation from radial conduction as wells interact at the symmetry boundary.

At large well spacings, the wells do not interact and the thermal response follows the same semi-infinite radial flow behavior observed for single isolated wells. At close well spacings, the thermal response approaches the response due to heat flow into a one dimensional semi-infinite space, with response scaling as:

$$T_g(y, t) = f(t^{0.5}) \quad (26)$$

Figure 8 shows a semi-logarithmic plot of the data. The response at large well separation distance shows a log-linear profile as expected for an isolated well. The deviation of the problem from a radial problem to a one-dimensional problem is shown in Fig. 9. The one-dimensional response is seen at low well spacing.

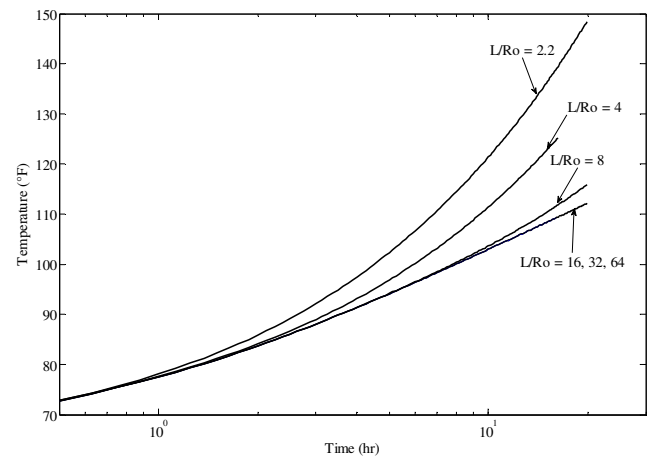


Fig. 8: Semi-logarithmic plot of the data with transition from radial semi-infinite solid conduction to one-dimensional semi-infinite solid conduction.

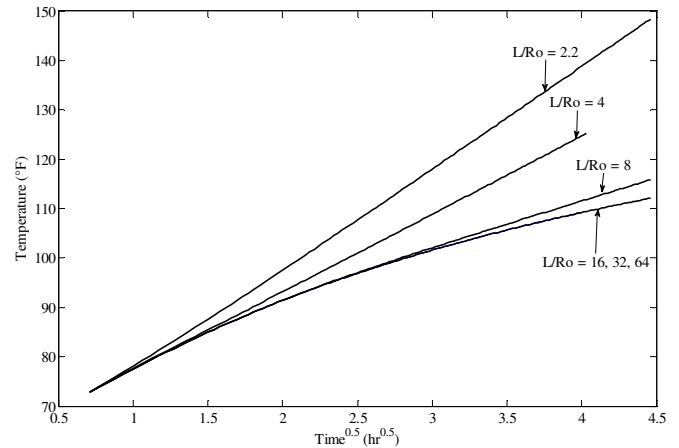


Fig. 9: Square root of time plot of the data showing the transition from radial semi-infinite solid conduction to one-dimensional semi-infinite solid conduction.

The slopes of the curves in Fig. 9 become linear due to thermal interaction with adjacent wells and differ depending on the well

spacing. An interesting note is the slopes of the plots in Fig. 9 are the tangent of the axisymmetric solution at the transition from axisymmetric to one-dimensional semi-infinite solid behavior. The time at which the plots deviate from the axisymmetric solution is calculated assuming a constant heat flux solution and is given by

$$\frac{4Fo}{e^\gamma} = \frac{4\alpha_g t_{prop}}{(0.5L - R_o)^2 e^\gamma} = 1 \quad (27)$$

where γ is Euler's constant.[9] The propagation time (t_{prop}) is when a temperature change occurs at the axis of symmetry between the wells. The plots in Fig. 9 illustrate propagation times of 0.206 hr for $L/R_o = 2.2$, 1.3 hr for $L/R_o = 4$, 7.1 hr for $L/R_o = 8$, and 32.5 hr for $L/R_o = 16$. Although these experiments and simulations primarily address the heating of geothermal wells, the same analysis applies when studying the cooling of geothermal wells. Assuming the energy input during the summer equals the energy extracted during the winter, the necessary distance between wells, L , is calculated by assuming that the heating season time, t_{prop} , is 3-6 months giving

$$L = 2R_o + 2\sqrt{\frac{4\alpha_g t_{prop}}{e^\gamma}} = 9.3 - 13.2m \quad (28)$$

Therefore, to provide the highest efficiency wells, the spacing should be larger than 9.3-13.2 meters (30-43ft). Common practice in geothermal well field spacing with ground properties similar to this study is currently 30-35ft.

CONCLUSIONS

Two detailed numerical models for a geothermal well were developed. The models depicted the thermal propagation through an open-loop geothermal well, and the ground formation. One model assessed the thermal effect of operating multiple wells in proximity to one another. Experimental data was used to verify the accuracy of the models. The following conclusions were derived:

1. Ground formations surrounding geothermal wells were modeled using "effective" properties with high accuracy.
2. An axisymmetric model was produced that encapsulated the transient nature of the problem particularly during heat up.
3. A multiple well model for an infinite line of wells was produced that demonstrated the conversion from radial conduction to one-dimensional conduction.
4. The common practice of spacing wells approximately 10 meters apart was confirmed.

An integral part of this problem that was not studied was the effect of the porosity of the ground formation surrounding the

well. Also, an important aspect of geothermal well operation is associated with "bleeding" the system, where porous flow is promoted by returning less water than is removed from the well. These parameters of the problem will be studied in future research.

ACKNOWLEDGMENTS

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NOMENCLATURE

A_{cd}	Cross-sectional area for flow down the annulus (m^2)
A_{cu}	Cross-sectional area for flow up the polypipe (m^2)
Bi	Biot number
Co_d	Courant number for annular flow
Co_u	Courant number for return pipe flow
C_{pw}	Heat capacity of water (J/kgK)
D_{ip}	Inner diameter of polypipe (m)
D_{op}	Outer diameter of polypipe (m)
Fo	Fourier number
H	Depth of the well (m)
L	Distance between wells (m)
NTU_{1d}	Number of transfer units in annular region to the ground formation
NTU_{1u}	Number of transfer units from the return pipe to the annular region
NTU_{2d}	Number of transfer units from the annular region to the return pipe
Nu_D	Nusselt number based on the diameter
P	Perimeter of the well (m)
Pr	Prandtl number of the water
Q	Heat load applied to the well (J/s)
Re_D	Reynolds number based on the diameter
R_o	Radius of the well (m)
R_{pipe}	Resistance through the polypipe (K/W)
T_{BOT}	Temperature of water at the bottom of the well ($^{\circ}C$)
T_g	Temperature of the ground formation ($^{\circ}C$)
T_{IN}	Temperature of water entering the well ($^{\circ}C$)
T_o	Initial ground formation temperature ($^{\circ}C$)
T_{OUT}	Temperature of water exiting the polypipe ($^{\circ}C$)
T_r	Temperature of the return flow through the polypipe ($^{\circ}C$)
T_w	Temperature of the flow down the annulus of the well ($^{\circ}C$)
$V_{annulus}$	Volume of water inside the annulus
V_{pipe}	Volume of the water inside the return pipe
c_f	Coefficient of friction between the water and a surface
h_o	Convective heat transfer coefficient for the well wall (W/m^2K)
h_{op}	Convective heat transfer coefficient on the outside wall of the polypipe (W/m^2K)

h_{ip}	Convective heat transfer coefficient on the inside wall of the polypipe (W/m ² K)
k_p	Conductivity of pipe (W/mK)
m	Water flow rate through the well (kg/s)
t_{heatup}	Time associated with heat up of the well using Courant number
t_{prop}	Time for thermal front to propagate to symmetric boundary with adjacent well
α_g	Thermal diffusivity of the ground (m ² /s)
ΔL_1	Length between the base node and an adjacent node
ΔL_2	Length between base node and adjacent node not equal to ΔL_1 due to non-uniform meshing
Δs_m	Time step for Duhamel superposition summation
λ_n	Characteristic value for Bessel function of the first kind of order zero
ρ_w	Density of water(kg/m ³)

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